

SUKHOBUZIMSKOYE, Russian Federation

WMO: 294770

Lat: 56.4971N

Lon: 93.2644E

Elev: 164

StdP: 99.37

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-44.6	-39.5	-46.5	0.0	-44.4	-42.4	0.1	-39.5	9.1	-10.6	8.3	-7.9	0.6	220	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	30.2	19.5	28.4	18.6	26.7	17.9	20.9	27.8	19.9	26.3	18.8	24.7	2.4	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	13.6	23.9	17.5	12.8	22.8	16.5	12.0	21.9	61.4	27.7	57.9	26.3	54.3	24.7	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.0	6.1	-44.5	33.6	4.1	1.5	-47.5	34.7	-49.9	35.5	-52.1	36.4	-55.1	37.4		
(5)			-43.4	22.1	4.5	0.9	-46.6	22.8	-49.3	23.3	-51.8	23.8	-55.1	24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.3	-20.8	-16.2	-7.3	3.1	9.3	17.0	18.9	16.5	8.4	0.4	-11.4	-15.1
	DBStd	15.23	11.37	8.64	7.78	4.45	5.10	4.32	2.95	3.73	4.48	5.37	9.77	10.11
	HDD10.0	4322	953	733	537	212	77	4	0	3	84	298	642	778
	HDD18.3	6679	1212	967	795	458	282	73	29	79	299	556	892	1037
	CDD10.0	787	0	0	0	3	55	213	276	204	35	1	0	0
	CDD18.3	103	0	0	0	0	2	33	46	22	0	0	0	0
	CDH23.3	1328	0	0	0	3	57	499	476	277	17	0	0	0
	CDH26.7	375	0	0	0	1	12	154	128	78	1	0	0	0
Wind	WSAvg	2.5	2.2	2.2	2.5	3.1	3.0	2.3	2.1	2.0	2.3	2.8	2.7	2.6
Precipitation	PrecAvg	385	13	12	12	22	37	50	49	68	43	30	26	24
	PrecMax	476	26	23	31	40	74	82	98	104	73	54	46	48
	PrecMin	291	3	5	2	5	8	18	3	40	10	8	12	11
	PrecStd	52	7	5	8	10	20	20	22	18	18	13	9	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	2.4	9.0	21.4	29.2	33.0	32.2	32.0	25.8	16.2	9.6	3.0
		MCWB	-1.2	-0.4	4.3	11.1	16.6	19.5	20.0	21.0	16.5	10.0	5.9	0.7
	2%	DB	-0.9	0.4	6.5	16.9	25.4	30.3	29.9	29.0	21.7	12.2	5.2	0.6
		MCWB	-2.8	-1.8	2.7	8.9	14.6	19.3	19.6	20.0	15.0	7.6	2.5	-1.1
	5%	DB	-3.0	-1.2	4.7	14.1	22.1	28.2	28.0	26.5	18.9	9.7	3.3	-1.0
		MCWB	-4.5	-3.0	1.4	7.8	12.7	18.3	18.7	18.2	13.8	6.5	1.2	-2.3
	10%	DB	-4.9	-3.2	3.0	11.7	19.6	26.4	26.1	24.2	16.2	7.7	1.1	-2.8
		MCWB	-6.1	-4.7	0.2	6.0	12.0	17.5	18.2	17.1	11.9	5.2	-0.4	-4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	-0.2	4.8	11.8	17.5	21.7	21.7	21.8	17.0	10.3	6.0	1.0
		MCDB	0.5	1.8	8.0	18.2	27.1	29.3	28.7	29.0	23.6	14.8	9.4	2.7
	2%	WB	-2.6	-1.5	3.0	9.8	15.2	20.7	20.7	20.4	15.6	8.3	2.7	-1.0
		MCDB	-0.9	0.2	5.9	16.5	22.9	28.2	26.9	27.2	21.1	11.3	4.8	0.4
	5%	WB	-4.4	-3.0	1.7	8.1	13.7	19.4	19.9	19.1	14.0	6.7	1.2	-2.3
		MCDB	-3.0	-1.2	4.3	13.4	21.0	26.4	26.1	24.5	18.3	9.3	3.1	-1.0
	10%	WB	-6.2	-4.8	0.5	6.5	12.3	18.2	19.0	17.9	12.2	5.2	-0.3	-3.9
		MCDB	-4.9	-3.2	2.9	11.2	18.7	24.8	24.6	23.0	15.9	7.5	1.1	-2.8
Mean Daily Temperature Range	5% DB	MDBR	9.3	11.0	11.5	10.5	11.7	12.7	11.0	11.4	10.2	8.0	8.1	8.8
		MCDBR	9.2	10.1	11.7	16.1	17.0	17.4	14.9	15.8	16.2	12.2	6.7	9.4
		MCWBR	8.1	8.5	8.6	9.4	8.5	7.5	6.2	7.3	9.6	8.1	5.0	8.3
	5% WB	MCDBR	8.6	9.6	10.3	14.7	14.8	14.8	12.6	13.6	15.5	11.4	6.6	9.0
		MCWBR	7.7	8.4	7.7	9.0	8.3	6.8	6.2	7.0	9.5	7.8	5.2	8.2
Clear-Sky Solar Irradiance	taub	0.287	0.319	0.363	0.395	0.415	0.415	0.438	0.415	0.370	0.351	0.319	0.285	
	taud	2.011	1.991	1.954	2.131	2.188	2.280	2.225	2.256	2.336	2.231	2.056	1.960	
	Ebn at Noon	557	699	766	814	822	828	797	789	775	670	508	437	
	Edh at Noon	65	100	134	132	133	123	128	116	93	79	62	52	
All-Sky Solar Radiation	RadAvg	0.75	1.73	3.09	4.32	4.92	5.96	5.36	4.25	2.68	1.39	0.67	0.43	
	RadStd	0.09	0.12	0.21	0.29	0.50	0.50	0.35	0.36	0.36	0.10	0.10	0.06	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (17 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page